Lab 14

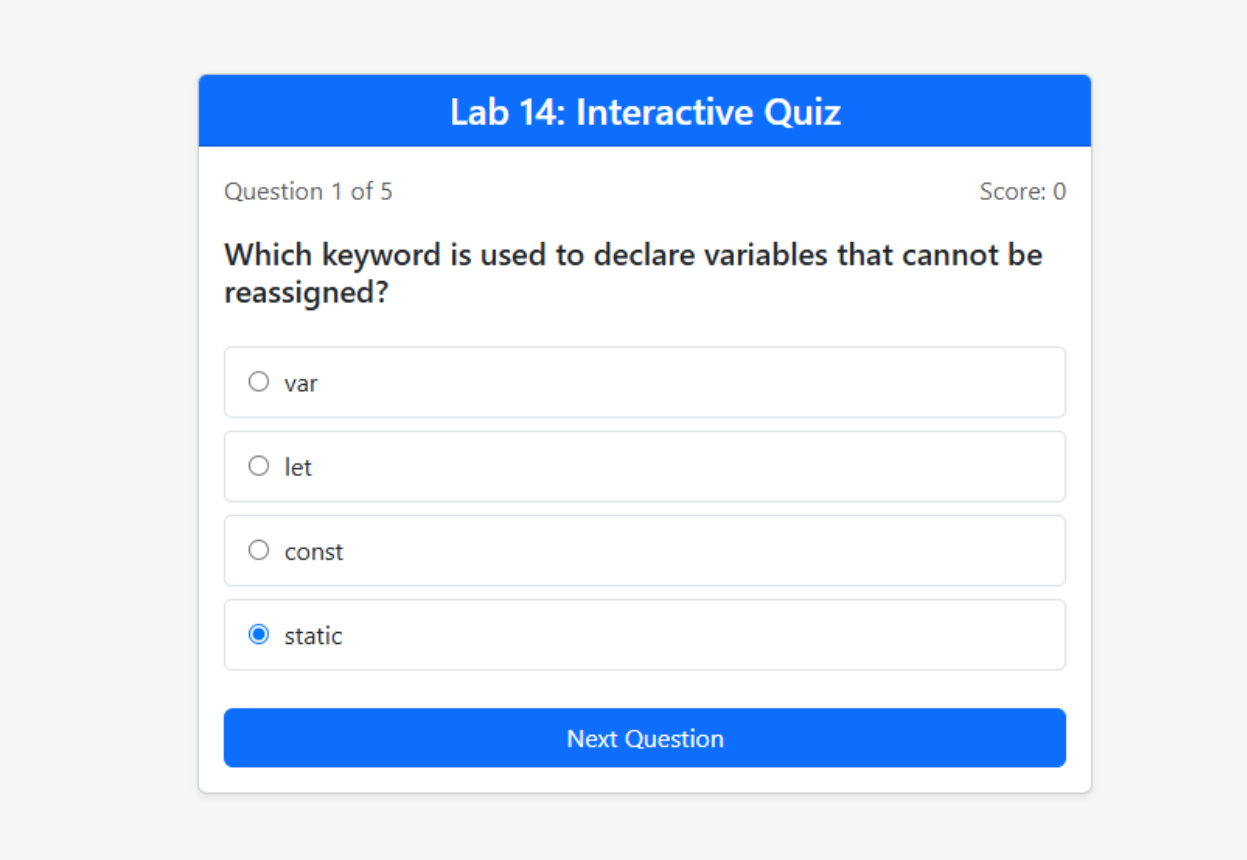
Q : Create a quiz application that displays one question at a time. Track the user's score and show the final result at the end.

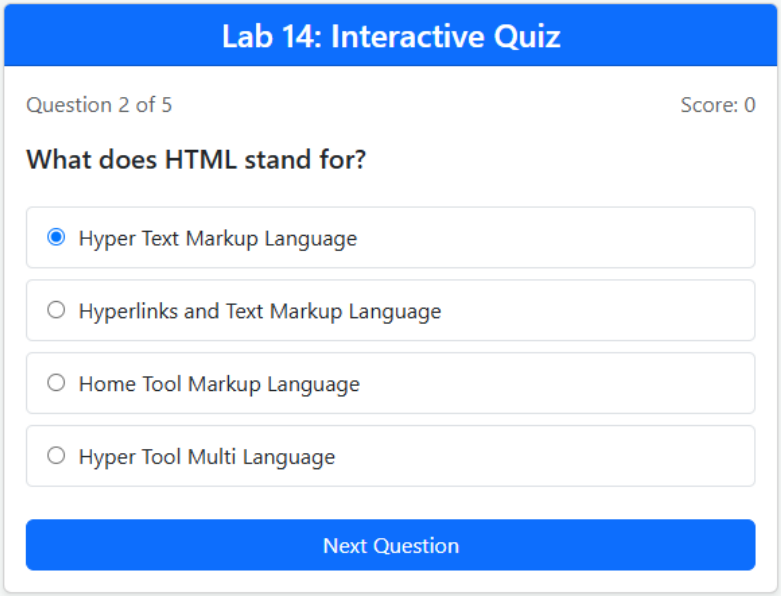
TAGS USED	ATTRIBUTES USED	DESCRIPTION
<!DOCTYPE html>	--	Declares the document as an HTML5 document.
<html>	lang	Defines the root element of the HTML document.
<head>	--	Contains metadata, title and stylesheet information.
<meta>	charset, name, content	Specifies character encoding and responsive viewport settings.
<title>	--	Defines the title displayed in the browser tab.
<link>	href, rel	Links the Bootstrap CSS stylesheet from the CDN.
<body>	class	Contains all visible webpage content and uses Bootstrap background styling.
<header>	class	Defines the header section and uses Bootstrap background, text, alignment and

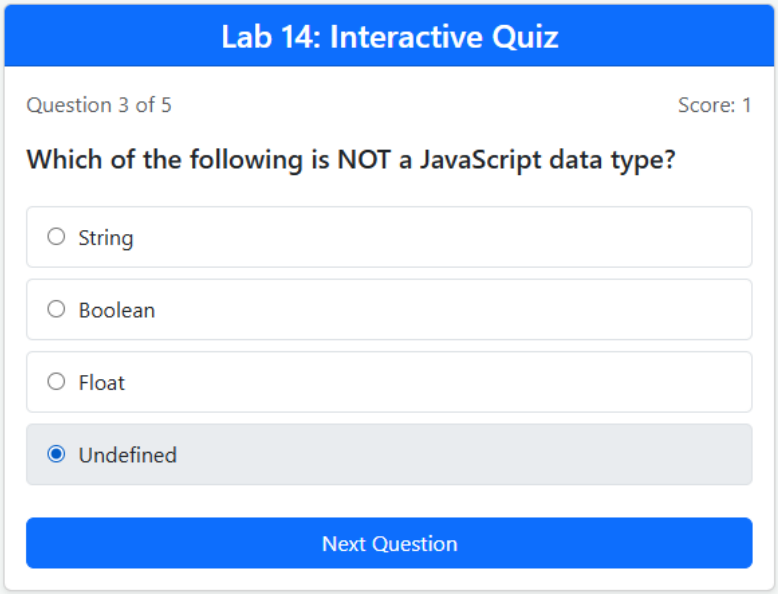
		padding classes.
——<img>	src, alt, class, style	Displays the Marwadi University logo and provides alternative text.
<h1>	class	Defines the main heading and uses Bootstrap font-weight styling.
<h2>	class	Defines the department heading and uses Bootstrap heading styling.
<nav>	class	Defines the navigation bar and uses Bootstrap background, alignment and padding classes.
<a>	href, class	Creates navigation links to different sections and applies Bootstrap link styling and spacing.
<section>	id, class	Defines separate webpage sections and uses Bootstrap container and margin classes.
	class	Creates Bootstrap containers, cards, card bodies, rows, columns and alert components.
<h3>	class	Defines section headings and uses

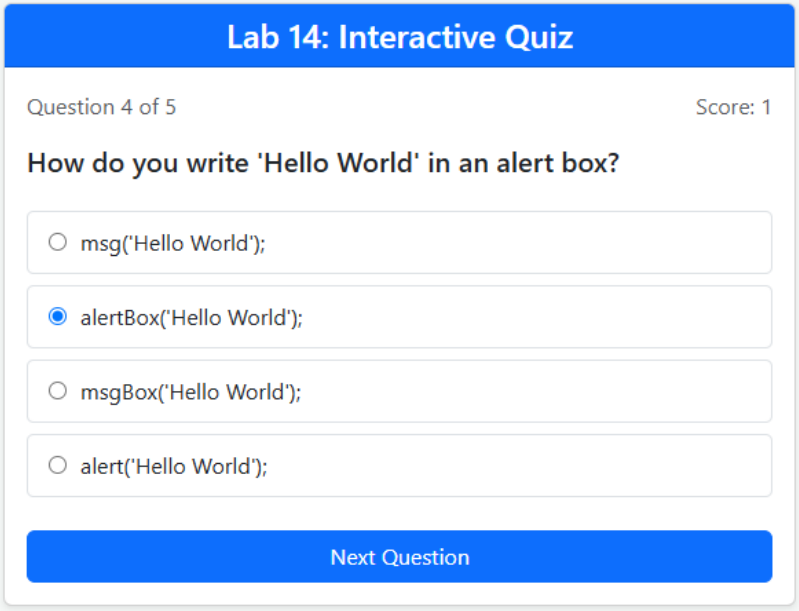
		Bootstrap text-color and font-weight classes.
<p>	class	Defines paragraphs and uses Bootstrap spacing classes where specified.
	class	Creates an unordered list and uses Bootstrap list-group styling.
	class	Creates an ordered list and uses Bootstrap numbered list-group styling.
	class	Defines list items and uses Bootstrap list-group-item styling.
<footer>	class	Defines the footer and uses Bootstrap background, text color, alignment and padding classes.

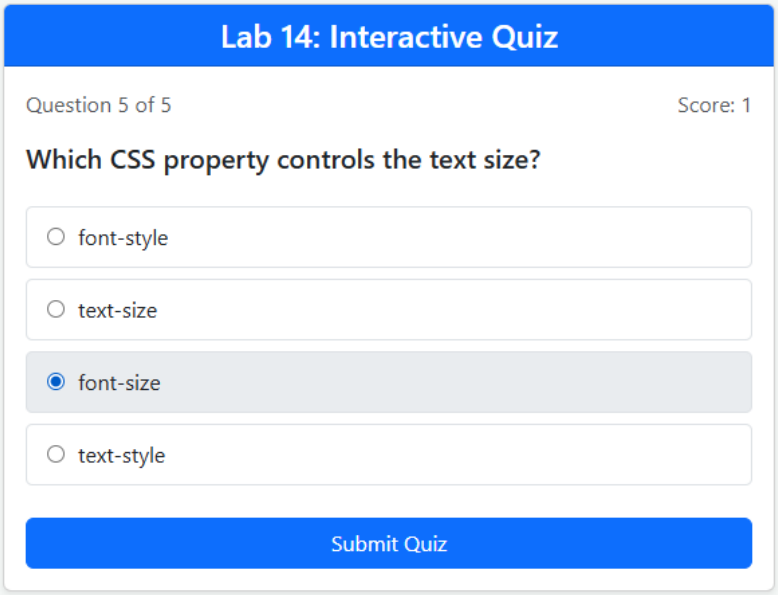
Output :

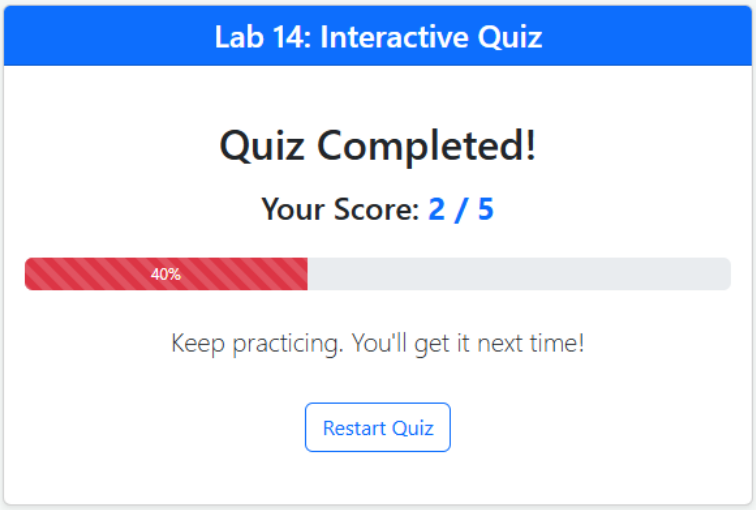

Lab 14: Interactive Quiz
Question 1 of 5
Score: 0
Which keyword is used to declare variables that cannot be reassigned?
var
let
const
static
Next Question


Lab 14: Interactive Quiz
Question 2 of 5
Score: 0
What does HTML stand for?
Hyper Text Markup Language
Hyperlinks and Text Markup Language
Home Tool Markup Language
Hyper Tool Multi Language
Next Question


Lab 14: Interactive Quiz
Question 3 of 5
Score: 1
Which of the following is NOT a JavaScript data type?
String
Boolean
Float
Undefined
Next Question


Lab 14: Interactive Quiz
Question 4 of 5
Score: 1
How do you write 'Hello World' in an alert box?
msg('Hello World');
alertBox('Hello World');
msgBox('Hello World');
alert('Hello World');
Next Question


Lab 14: Interactive Quiz
Question 5 of 5
Score: 1
Which CSS property controls the text size?
font-style
text-size
font-size
text-style
Submit Quiz


Lab 14: Interactive Quiz
Quiz Completed!
Your Score: 2 / 5
40%
Keep practicing. You'll get it next time!
Restart Quiz

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Interactive Quiz Application</title>
   <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
   <style>
      body { background-color: #f4f7f6; }
      .quiz-container { max-width: 600px; margin-top: 50px; }
      .option-label {
         display: block;
         padding: 10px 15px;
         border: 1px solid #dee2e6;
         border-radius: 5px;
         cursor: pointer;
         transition: background-color 0.2s;
      }
      .option-label:hover { background-color: #e9ecef; }
      input[type="radio"] { margin-right: 10px; }
      .hidden { display: none !important; }
   </style>
</head>
<body>

<div class="container quiz-container">
```

```
  <div class="card shadow-sm">
    <div class="card-header bg-primary text-white text-center">
      <h4 class="mb-0" id="quiz-title">Lab 14: Interactive Quiz</h4>
    </div>
    <div class="card-body">

      <!-- Quiz Screen -->
      <div id="quiz-screen">s
        <div class="d-flex justify-content-between mb-3 text-muted">
          <span id="question-progress">Question 1 of 5</span>
          <span id="score-tracker">Score: 0</span>
        </div>
        <h5 id="question-text" class="mb-4">Question goes here?</h5>

        <div id="options-container" class="mb-4">
          <!-- Options injected via JavaScript -->
        </div>

        <button id="next-btn" class="btn btn-primary w-100" disabled>Next
Question</button>
      </div>

      <!-- Result Screen -->
      <div id="result-screen" class="hidden text-center py-4">
        <h2 class="mb-3">Quiz Completed!</h2>
        <h4 class="mb-4">Your Score: <span id="final-score" class="text-primary
fw-bold">0 / 0</span></h4>
        <div class="progress mb-4" style="height: 25px;">
          <div id="score-bar" class="progress-bar progress-bar-striped progress-
bar-animated" role="progressbar" style="width: 0%;" aria-valuenow="0" aria-
valuemin="0" aria-valuemax="100">0%</div>
        </div>
        <p id="feedback-message" class="lead"></p>
```

```
            <button id="restart-btn" class="btn btn-outline-primary mt-3">Restart
Quiz</button>
        </div>

    </div>
  </div>
</div>

<script>
  const quizData = [
    {
      question: "Which keyword is used to declare variables that cannot be
reassigned?",
      options: ["var", "let", "const", "static"],
      correct: 2
    },
    {
      question: "What does HTML stand for?",
      options: ["Hyper Text Markup Language", "Hyperlinks and Text Markup
Language", "Home Tool Markup Language", "Hyper Tool Multi Language"],
      correct: 0
    },
    {
      question: "Which of the following is NOT a JavaScript data type?",
      options: ["String", "Boolean", "Float", "Undefined"],
      correct: 2
    },
    {
      question: "How do you write 'Hello World' in an alert box?",
      options: ["msg('Hello World');", "alertBox('Hello World');", "msgBox('Hello
World');", "alert('Hello World');"],
      correct: 3
    },
```

```
    {
      question: "Which CSS property controls the text size?",
      options: ["font-style", "text-size", "font-size", "text-style"],
      correct: 2
    }
  ];

  let currentQuestionIndex = 0;
  let score = 0;

  const quizScreen = document.getElementById('quiz-screen');
  const resultScreen = document.getElementById('result-screen');
  const questionText = document.getElementById('question-text');
  const optionsContainer = document.getElementById('options-container');
  const nextBtn = document.getElementById('next-btn');
  const questionProgress = document.getElementById('question-progress');
  const scoreTracker = document.getElementById('score-tracker');

  function loadQuestion() {
    nextBtn.disabled = true;
    const currentQ = quizData[currentQuestionIndex];
    questionText.textContent = currentQ.question;
    questionProgress.textContent = `Question ${currentQuestionIndex + 1} of
${quizData.length}`;
    scoreTracker.textContent = `Score: ${score}`;

    optionsContainer.innerHTML = '';
    currentQ.options.forEach((option, index) => {
      const wrapper = document.createElement('div');
      wrapper.className = 'mb-2';

      const label = document.createElement('label');
      label.className = 'option-label';
```

```
        const radio = document.createElement('input');
        radio.type = 'radio';
        radio.name = 'quiz-option';
        radio.value = index;
        radio.addEventListener('change', () => nextBtn.disabled = false);

        const textNode = document.createTextNode(option);

        label.appendChild(radio);
        label.appendChild(textNode);
        wrapper.appendChild(label);
        optionsContainer.appendChild(wrapper);
      });

      if (currentQuestionIndex === quizData.length - 1) {
        nextBtn.textContent = 'Submit Quiz';
      } else {
        nextBtn.textContent = 'Next Question';
      }
    }

    function checkAnswer() {
      const selectedOption = document.querySelector('input[name="quiz-
option"]:checked');
      if (selectedOption) {
        const answer = parseInt(selectedOption.value);
        if (answer === quizData[currentQuestionIndex].correct) {
          score++;
        }
      }
    }
```

```
function showResults() {
    quizScreen.classList.add('hidden');
    resultScreen.classList.remove('hidden');

    const finalScoreText = document.getElementById('final-score');
    const scoreBar = document.getElementById('score-bar');
    const feedbackMessage = document.getElementById('feedback-message');

    finalScoreText.textContent = `${score} / ${quizData.length}`;

    const percentage = Math.round((score / quizData.length) * 100);
    scoreBar.style.width = `${percentage}%`;
    scoreBar.textContent = `${percentage}%`;
    scoreBar.setAttribute('aria-valuenow', percentage);

    if (percentage === 100) {
        scoreBar.classList.add('bg-success');
        feedbackMessage.textContent = "Perfect! Excellent job.";
    } else if (percentage >= 60) {
        scoreBar.classList.add('bg-info');
        feedbackMessage.textContent = "Good effort! You passed.";
    } else {
        scoreBar.classList.add('bg-danger');
        feedbackMessage.textContent = "Keep practicing. You'll get it next time!";
    }
}

nextBtn.addEventListener('click', () => {
    checkAnswer();
    currentQuestionIndex++;

    if (currentQuestionIndex < quizData.length) {
        loadQuestion();
```

```
        } else {
            showResults();
        }
    });

    document.getElementById('restart-btn').addEventListener('click', () => {
        currentQuestionIndex = 0;
        score = 0;
        quizScreen.classList.remove('hidden');
        resultScreen.classList.add('hidden');
        document.getElementById('score-bar').className = 'progress-bar progress-
bar-striped progress-bar-animated';
        loadQuestion();
    });

    // Initialize Quiz
    loadQuestion();
</script>
</body>
</html>
```